FusionAgent: A Multimodal Agent with Dynamic Model Selection for Human Recognition

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Abstract

Systematic human recognition requires integrating multiple biometric traits such as face, gait, and body shape, through specialized models to achieve robustness in unconstrained scenarios. However, existing score-fusion strategies typically adopt a static design, combining all models for every test sample regardless of sample quality. This not only increases unnecessary computation but can degrade performance by incorporating noisy or unreliable modalities. To overcome these limitations, we propose ***Fusion-Agent****, a novel agentic framework that leverages a Multimodal Large Language Model (MLLM) to perform dynamic, sample-specific model selection. Each model is treated as a tool, and through Reinforcement Fine-Tuning (RFT) with a metric-based reward, the agent learns to adaptively determine the optimal model combination for each test input. To address the model score misalignment and embedding heterogeneity, we introduce Anchor-based Confidence Top-k (ACT) score-fusion, which anchors on the most confident model and integrates complementary predictions in a confidence-aware manner. Extensive experiments on multiple whole-body biometric benchmarks demonstrate that FusionAgent significantly outperforms SoTA methods, underscoring the critical role of dynamic, explainable, and robust model fusion in real-world recognition systems. The proposed framework is scalable and adaptable to a wide range of multi-modal and multi-model tasks, such as vision-language retrieval, indicating its potential relevance to broader application scenarios. The code and model will be publicly released upon publication.*

1. Introduction

Human (whole-body) recognition is a challenging task that leverages multimodal traits (*e.g.*, face, gait, and body shape) and multiple specialized models to identify individuals accurately [43]. In contrast to traditional unimodal biometric systems that rely on a single modality, such as face recognition (FR) [6, 13, 14], gait recognition (GR) [39, 42], or person re-identification (ReID) [7, 17, 33], systematic whole-body recognition integrates complementary information from diverse modalities to improve the recognition performance. This fusion enhances robustness but introduces challenges in effectively and efficiently combining multi-model outputs. Such integration is critical to overcoming the limitations of unimodal systems; for instance, FR models struggle with occluded faces, where GR or ReID models can provide reliable alternatives. These capabilities are vital in real-world surveillance applications, which demand high accuracy and robustness despite low-quality imagery [16].

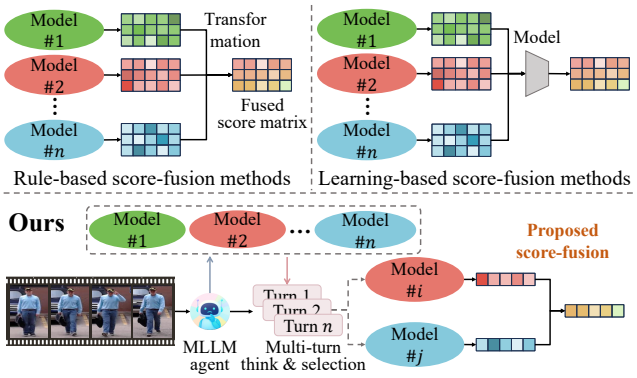

Figure 1. **Comparison of score-fusion methods. Top:** Rule-based methods apply predefined transformations to fuse all model scores, while learning-based methods infer a fusion model from data but still assume that every model contributes to all test samples. **Bottom:** our framework leverages an MLLM agent to dynamically select a subset of models, followed by the proposed score-fusion strategy, enabling adaptive and robust integration.

Score-level fusion is essential in human recognition systems. As shown in Fig. 1, existing methods fall into two main categories: rule-based and learning-based. Empirical approaches use predefined rules to combine scores [11, 31], making them computationally efficient and training-free. In contrast, learning-based methods learn fusion strategies from data [22, 34, 43], capturing complex relationships at the cost of requiring training data and model design. A key

limitation of both approaches is their inherent assumption that all models provide complementary information, leading them to treat the model combination as fixed for every input, even those from low-quality or unreliable modalities. For instance, in cases where a video frame only captures the back view of a person, incorporating FR models would be inappropriate. Although some methods, such as QME [43], attempt to mitigate poor input quality within the fusion process, they still allow low content inputs to influence the output. This suggests that the conventional fusion designs may be suboptimal. An ideal fusion framework should instead: 1) dynamically select the most suitable model combination for each input, and 2) effectively integrate the outputs from the selected models. To systematically explore the potential of such an approach, we focus on the following question:

> *How can we adaptively choose the **optimal model combination for each test sample** while rigorously testing whether such selection improves performance even with an **ad hoc** score-fusion algorithm?*

To address the question, we introduce a unified framework centered around two core components, each designed to systematically respond to model selections and score-fusion methods. In response to model selections, we propose **FusionAgent**, an MLLM-based agent that performs dynamic model selection per sample. Each biometric model is wrapped as a tool, providing the agent with a score vector and predicted label. Through Reinforcement Fine-Tuning (RFT) guided by the **proposed metric-based reward**, the agent learns to identify sample-specific model combinations by analyzing query patterns, effectively answering how optimal ensembles can be found for each input. To explore score-fusion methods, we design a lightweight **Anchor-based Confidence Top-k score-fusion method (ACT)**. This method leverages the most confident model selected by the agent, then integrates only the top-matching scores scaled by confidence weights. By doing so, ACT mitigates misalignment from sample-wise model selection and embedding heterogeneity, demonstrating that even a simple fusion rule can yield strong performance when guided by adaptive model selection. The proposed method can be extended to various retrieval-oriented scenarios. For instance, it could facilitate cross-modal retrieval between visual and textual data or enhance domain-specific search systems. Our main contributions are as follows:

- We propose FusionAgent, an agentic framework that leverages an MLLM to perform explainable, sample-wise dynamic model selection.
- We introduce an Anchor-based Confidence Top-k score-fusion method (ACT) to mitigate score misalignment arising from dynamic model selection, ensuring robust integration of heterogeneous score outputs.
- We design a metric-based reward function, central to our optimization, which directly aligns the agent's selection strategy with final performance metrics.
- Extensive experiments on multiple whole-body biometric benchmarks demonstrate the superiority of our approach over state-of-the-art methods, even using conventional score-fusion methods.

2. Related Work

2.1. Whole-body Biometric Recognition

Whole-body biometric recognition systems combine detectors (*e.g.*, for whole body and face), embedding models, and fusion modules to leverage multi-modal cues such as face and body features [5, 16, 18]. The key challenge is to effectively integrate the complementary strengths of different modalities and their dedicated models to maximize overall performance. For example, FR models excel with high-quality frontal faces but struggle under adverse conditions like oblique angles [13, 14]. In contrast, GR models focus on clothing-invariant dynamic body attributes [39, 42], while ReID models take a holistic approach to extract comprehensive appearance features [7, 17, 36, 37]. Prior fusion methods typically use all available models for every query [3, 16, 18, 22, 34, 43], ignoring the sample-dependent nature of optimal model combinations. A clear case is a low-resolution image with a side-view of a person: FR models may be unreliable, making GR or ReID models more critical for that specific sample. Therefore, we propose FusionAgent to dynamically select an optimal model combination for each test sample through explainable analysis, thereby tailoring the fusion process to individual inputs.

2.2. Score-fusion

Score-level fusion integrates similarity scores from multiple modalities to improve recognition performance [30]. This approach is broadly categorized into two paradigms: rule-based and learning-based methods. Rule-based methods employ fixed rules—such as Z-score, Min-max normalization, max/min fusion [11, 12, 26, 40], and likelihood ratio-based fusion [8, 21, 23, 24, 35], offering simplicity and efficiency. In contrast, learning-based methods optimize fusion during training [3, 22, 34]. These include recent quality-aware approaches like QME [43], which performs weighted score-fusion based on input quality. In contrast, we introduce a simple score-fusion algorithm with an anchor. This design, combined with selective model input, leads to superior performance compared with existing fusion strategies.

2.3. Agents and Reinforcement Learning

The advent of tool-augmented MLLMs has spurred the development of agentic systems: models capable of planning, reasoning, and interacting with external tools to solve com-

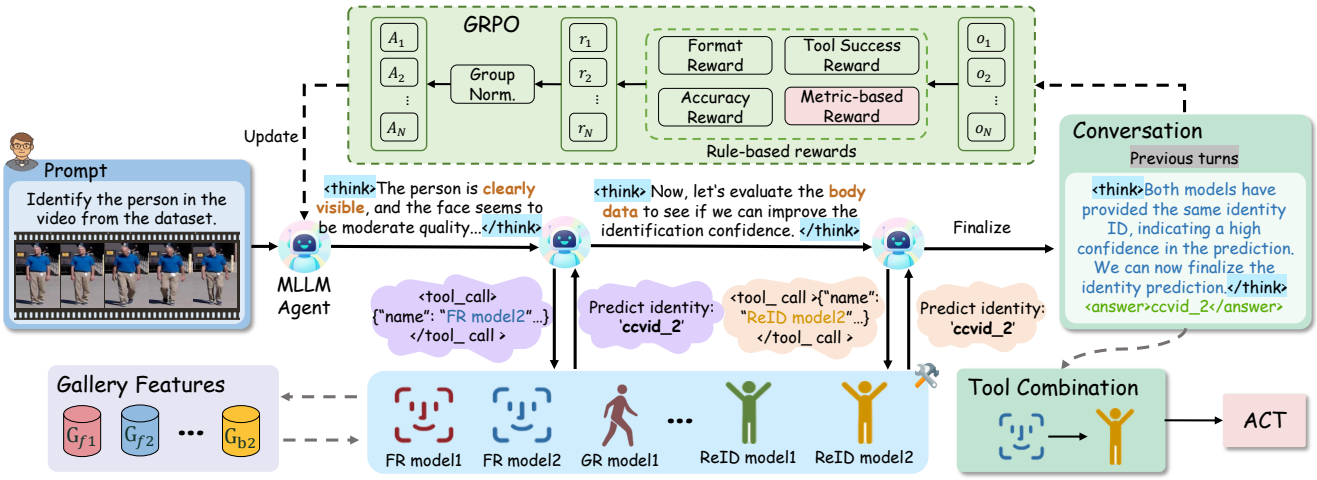

Figure 2. **Overview of the FusionAgent framework.** Recognition models are wrapped as tools to generate score vectors and predicted identities based on gallery features. The MLLM agent receives multimodal biometric inputs and performs a reasoning-action step through multi-turn, selectively invokes tools, and integrates predictions into a final identity decision and fused score vector. The agent is optimized with reinforcement fine-tuning using rule-based rewards, including the proposed metric-based reward.

plex tasks. Recent studies [2, 20, 38, 44] have equipped MLLMs with tools to address diverse challenges such as visual grounding, image super-resolution, and advanced visual understanding. Furthermore, subsequent research has employed RFT with rule-based rewards, such as Group Relative Policy Optimization (GRPO) [27], which offers improved generalizability with lower data dependency [19, 28, 41]. Building upon these advancements, we further propose a metric-based reward to optimize model selections of FusionAgent, effectively capturing data-dependent patterns.

3. Proposed Method

Problem Formulation. In biometric recognition, a query (or probe) is a sample sequence that is to be identified (1:N comparisons) or verified by comparison (1:1 comparison) with a gallery of previously enrolled subjects in the system. Let Q denote the set of query videos and G be the gallery. M is a predefined model set $\{m_1, m_2, \ldots, m_Z\}$ that is used for feature extraction. For a query $q \in Q$ and a model $m \in M$ (from a predefined model set M), m extracts features from q and produces a similarity score vector $\mathbf{s}_{m,q} \in \mathbb{R}^{|G|}$ with respect to each gallery item $g \in G$. The human recognition pipeline comprises two stages: (i) *sample-level model subset selection* and (ii) *query-based score fusion of selected models*. For each q, we seek an optimal subset of models $\mathbf{M}_q \subseteq M$ that maximizes performance. Given $\mathbf{M}_q$, we stack the per-model scores into a score matrix $\mathbf{S}_q \in \mathbb{R}^{|M_q| \times |G|}$ whose i-th row is $\mathbf{s}_{m_i,q}$. A score-fusion function f then maps $\mathbf{S}_q$ to a fused score vector $\mathbf{s}'_q = f(\mathbf{S}_q) \in \mathbb{R}^{|G|}$. We detail the model-selection procedure using an agent in Sec. 3.1, the supervision signals (reward functions) in Sec. 3.2, and the proposed score-fusion method in Sec. 3.3.

3.1. Agentic Training Framework

Training the agent presents two challenges: (i) the lack of ground-truth dialogue data for Supervised Fine-tuning (SFT), since grid searching optimal model combinations per sample is infeasible; and (ii) the intractability of dataset-level grid search for the best model combination per sample due to exponential growth in search space with models and samples. We thus employ GRPO, which enables trial-based learning without pre-labeled data. Additionally, we design a metric-based reward function that promotes exploration of diverse model combinations and facilitates implicit learning of the relationship between input characteristics and model performance (details in Sec. 3.2).

The overall framework is depicted in Fig. 2. For a given query q, the agent first analyzes its content and selects an initial model $a_q \in M_q$. For example, FR model 2 is selected as a clearly visible face is detected. Upon a successful function call, the tool returns $s_{a_q}(q)$ and the predicted label. Since score vectors are not directly interpretable by the agent, the predicted identity label is returned explicitly, while $s_{a_q}(q)$ is retained internally. The agent then decides whether to invoke another model or output its decision. This sequential decision-making process enables the flexible maximization of a long-term objective. [29, 38]. If the agent decides to output its decision, it also provides a reasoning summary that justifies the model selections. Reasoning at each step is necessary to enhance transparency and decision traceability. The whole conversation o_i, including the successful model selections, is then used to measure the

overall rewards. In GRPO, we sample N rollouts (*i.e.*, responses) for each query and jointly compute their advantages $\{A_1, A_2, \ldots, A_N\}$ from the corresponding rewards $\{r_1, r_2, \ldots, r_N\}$ to update the agent. Details of GRPO are provided in the supplementary.

Multi-turn Design. We adopt a ReAct-style (reason-before-act) multi-turn controller for tool use, rather than a single-shot plan generator [38]. This design is motivated by the need to decompose the complex task of multi-tool usage into a sequence of simpler, more manageable decisions. A single-turn approach would require the agent to generate a complete and static execution plan at once, a task plagued by a combinatorial action space and an inability to handle unexpected outcomes. This interleaving of reasoning with actions decomposes multi-tool execution into atomic steps and yields the following benefits:

i) Simplified Learning: Reduces the vast action space of generating a full plan to a single decision at each step, making the policy significantly more tractable to learn.

ii) Dynamic Adaptation: Allows the agent to observe tool outputs and adjust its strategy in real time, enabling error correction and flexible reasoning. These sequential actions also support effective credit assignment during inference.

3.2. Reward Functions

Reward functions play a central role in RFT-based agent training. Unlike metric learning or SFT, which minimize loss functions, RFT seeks to maximize reward. We design four reward functions: format reward, tool success reward, answer accuracy reward, and metric-based reward.

Format Reward. The agent is trained to produce structured responses that separate reasoning, tool calls, and final answers [19, 20, 27]. We adapt this to a multi-turn setting where each turn must be a structured format, such as *<think>...</think> <answer>...</answer>*. The reward is computed per turn, and the overall reward is averaged across all turns.

Tool Success Reward. This reward assesses whether the agent's tool calls are executable and yield valid results. Each tool call receives a binary success/failure score, and the reward is defined as the success rate across all tool calls in a trajectory. This incentivizes the agent to produce syntactically correct tool inputs.

Answer Accuracy Reward. This reward measures the correctness of the agent's final prediction a_i extracted from the answer tag *<answer>...</answer>* of the response o_i against the ground truth label y. Its primary objective is to prioritize factual accuracy over procedural correctness, which enables the agent to implicitly assess model reliability, especially when predictions conflict:

$$R_{acc}(a_i, y) = \begin{cases} 1, & \text{if } a_i = y, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Metric-based Reward. The metric-based reward guides the agent toward effective and dynamic model selection. A key challenge is to encourage sufficient exploration of diverse model combinations without prior knowledge of their optimality. To address this, the reward is computed based on the performance of the agent-selected model combination on the training set, using the score-fusion method from Sec. 3.3. The combination is derived from successfully executed conversations per sample.

However, exhaustively searching for the ground-truth optimal model combination for each sample is computationally infeasible. To promote exploration, we construct an augmented model selection $\mathbf{M_Q} \in \mathbb{R}^{|Q|\times|M|}$ based on the model combination M_{o_i} of the response o_i. Parameter $\gamma \in [0, 1]$ is the ratio of samples in $\mathbf{M_Q}$ whose model combination is the same as M_{o_i}. For the remaining $(1-\gamma)$, the mask entries are sampled from a Bernoulli distribution to diversify the model combinations explored. The column corresponding to the anchor model is always set to true, ensuring its consistent inclusion.

We apply the ACT (details in Sec. 3.3) based on $\mathbf{M_Q}$. The performance of the resulting fused scores matrix $\mathbf{S}'_{o_i}$ is then evaluated across the entire training set using four key metrics: True Accept Rate at a False Acceptance Rate (TAR@FAR), mean Average Precision (mAP), Rank-1 accuracy, and False Non-Identity Rate at a specified False Positive Identification Rate (FNIR@FPIR). The final metric-based reward R_{mat} is a composite score formulated to provide a holistic performance signal:

$$\mathbf{S}'_{o_i} = \text{ACT}(\mathbf{M_Q}, k), \tag{2}$$

$$\begin{aligned} R_{mat}(o_i) = &\text{Rank}(\mathbf{S}'_{o_i}) + \text{mAP}(\mathbf{S}'_{o_i}) \\ &+\text{TAR}(\mathbf{S}'_{o_i}) - \text{FNIR}(\mathbf{S}'_{o_i}). \end{aligned} \tag{3}$$

This formulation naturally aligns with both our optimization objective and real-world deployment needs. It rewards the agent for achieving higher accuracy and retrieval performance (TAR, mAP, Rank) while penalizing missed identifications (FNIR), thereby capturing operational performance requirements in a comprehensive manner. By consolidating these diverse metrics into a single scalar, the agent receives a clear and reliable training signal, effectively guiding its policy toward discovering superior model combinations.

3.3. Anchor-based Confidence Top-k Score-fusion

The proposed ACT (Anchor-based Confidence Top-k) score-fusion approach aims to dynamically and effectively combine scores from multiple models for a given query.

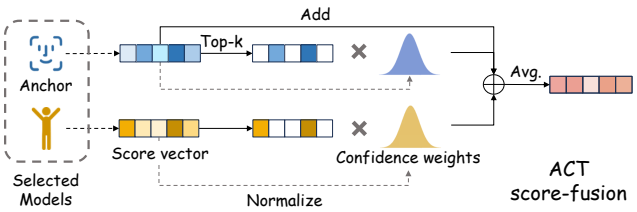

Figure 3. **Overview of the ACT score-fusion.** Based on tool execution results (*i.e.*, score vectors) and selected model combination, the first selected model serves as the anchor, and ACT produces the final score vector via confidence weighting and top-k filtering.

The overview is shown in the Fig. 3. Our approach is built on the principle of leveraging a stable and powerful "anchor model" $m_a \in \mathbf{M}_q$ to provide a robust score vector, while selectively incorporating normalized, high-confidence scores from the set of models. Let m_a be the first selected model in $\mathbf{M}_q$, and $s_{m,q,g}$ denote the similarity score between q and g for model m.

We begin by computing a contribution score $c_{m,q,g}$. This score is designed to leverage the most confident predictions while filtering out potential noise from low-scoring candidates, which is achieved through a Top-k selection process. Formally, for a score $c_{m,q,g}$ in $\mathbf{c}_{m,q} \in \mathbb{R}^{|G|}$, the contribution score is defined as:

$$c_{m,q,g} = \begin{cases} z_{m,q,g} \cdot s_{m,q,g} & \text{if } g \in \mathcal{T}_{m,q}, \\ 0 & \text{otherwise.} \end{cases} \tag{4}$$

where $\mathcal{T}_{m,q}$ is the set of indices of the k highest-scoring gallery items for query q from model m. The term $z_{m,q,g}$ represents the Z-score normalized score from m between q and g. This standardization makes the scores from different models with potentially different scales comparable. It is calculated as:

$$z_{m,q,g} = \frac{s_{m,q,g} - \mu_{m,q}}{\sigma_{m,q}}, \tag{5}$$

where $\mu_{m,q}$ and $\sigma_{m,q}$ are the mean and standard deviation of $\mathbf{s}_{m,q}$. The final fused score vector $\mathbf{s}'_q$ is computed via:

$$\mathbf{s}'_q = \frac{1}{1 + |\mathbf{M}_q|} \left(\mathbf{s}_{m_a,q} + \sum_{m \in \mathbf{M}_q} \mathbf{c}_{m,q} \right). \tag{6}$$

The term $(1 + |M_q|)$ serves as the normalization factor, ensuring a balanced contribution from all models. This strategy balances robust, general performance with specialized, high-confidence insights. The anchor model provides a more substantial contribution by establishing a global ranking structure, as its scores are applied unconditionally to all candidates. In contrast, the selected models provide sparse, localized refinements only for their top-k predictions. Fig. 4 provides a toy example of ACT score-fusion.

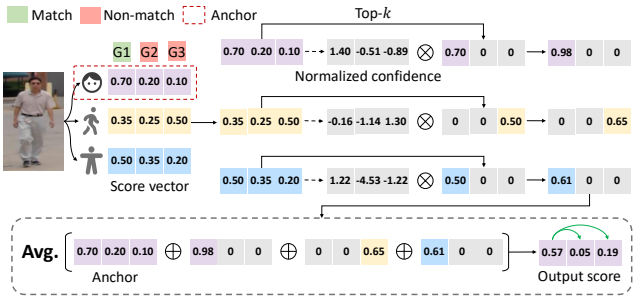

Figure 4. **A toy example of the proposed ACT score-fusion.** Three models are used with the FR model as the anchor and $k = 1$. ACT amplifies the gap between match and non-match scores through confidence-based top-k and anchor weighting, improving verification and open-set search performance.

4. Experiments

Datasets and Evaluation Protocol. We evaluate on popular human recognition datasets: CCVID [7], MEVID [4], and LTCC [25]. They comprise multi-view captures and cross-modal biometric data with diverse resolutions and temporal dynamics. Our evaluation protocol follows [43] for fair comparison. Performance is measured via standard metrics in general settings, *i.e.*, Rank-1 accuracy, mean Average Precision (mAP), verification rate (TAR@1.0%FAR), and open-set search metric (FNIR@1.0%FPIR), which collectively reflect real-world deployment requirements. For open-set search evaluation, we follow the protocol of [32, 43] to construct 10 random subsets of gallery subjects. Each subset, containing ∼20% of the subjects in the test set, serves as a non-mated list. We report the mean and standard deviation for the 10 trials.

Baselines. Our baselines include the following methods: Min/Max score-fusion [11], Z-score and min-max normalization [31], RHE [8], Farsight [16], and learning-based methods like BSSF [34], Weighted-sum [22], AsymA-O1 [9], and QME [43] (ICCV'25). We follow [43] to construct the biometric model pool (details provided in the Supp.). We also compare with the SoTA multimodal human recognition model [15] (CVPR'25).

Implementation Details. Following [43], We extract the center embeddings of subjects for training and gallery embeddings for testing by applying average pooling. Considering the demand for faster responses in practical applications, FusionAgent is based on Qwen2.5-VL-3B [1] and fine-tuned via GRPO [27]. For efficiency, we set a turn limit of 4 and train for 200 steps. All biometric models are frozen in training. For GRPO, we set $N = 6$ and a KL coefficient $\beta = 0.04$. LoRA [10] is applied with rank $r = 64$ and scaling factor $\alpha = 128$. The learning rate follows a linear decay schedule starting from 2×10^{-5}. The metric-based reward uses $\gamma = 0.8$. Based on training-set performance, we set

Method	*Comb.*	Rank1↑	mAP↑	TAR↑	FNIR↓
*AdaFace** [13]	♦	94.0	87.9	75.7	13.0 ± 3.5
CAL [7]	♠	81.4	74.7	66.3	52.8 ± 13.3
*BigGait** [39]	♣	76.7	61.0	49.7	71.1 ± 6.1
SapiensID [15]	●	92.6	77.8	-	-
Min-Fusion [11]		87.1	79.2	62.4	48.5 ± 8.7
Max-Fusion [11]		89.9	89.3	73.4	23.0 ± 10.1
Z-score [31]		92.2	90.6	73.9	15.1 ± 1.5
Min-max [31]		91.8	90.9	73.9	15.4 ± 2.5
Weigthed-sum [22]		91.7	90.6	73.6	15.4 ± 1.8
Asym-AO1 [9]	♦♠♣	92.3	90.0	74.0	15.9 ± 1.7
BSSF [34]		91.8	91.1	73.9	14.1 ± 1.3
Farsight [16]		92.0	91.2	73.9	13.9 ± 1.1
QME [43]		**94.1**	90.8	76.2	12.3 ± 1.4
FusionAgent (DA)		92.8	92.2	85.8	10.5 ± 1.5
FusionAgent (CoT)		93.4	**92.6**	**85.9**	$\mathbf{10.1 \pm 1.5}$

Table 1. **Performance on CCVID.** [Keys: **Best** and second best performance; *Comb.*: model combination; *: zero-shot performance; ♦: AdaFace for face modality; ♣: BigGait for gait modality; ♠: CAL of body modality; ●: SapiensID for face and body modalities; TAR: TAR@1%FAR; FNIR: FNIR@1%FPIR.]

Method	*Comb.*	Rank1↑	mAP↑	TAR↑	FNIR↓
*AdaFace** [13]	♦	18.5	5.9	2.4	99.8 ± 0.2
CAL [7]	♠	74.4	40.6	36.7	59.7 ± 7.3
AIM [37]	■	74.8	40.9	37.0	66.2 ± 7.5
SapiensID [15]	●	72.0	34.6	-	-
Min-Fusion [11]		38.1	13.5	12.4	81.9 ± 6.0
Max-Fusion [11]		62.5	33.3	16.8	94.8 ± 4.7
Z-score [31]		73.0	37.5	30.4	68.7 ± 9.2
Min-max [31]		73.2	38.1	31.9	75.1 ± 9.2
Weigthed-sum [22]	♦♠■	73.2	37.8	31.3	72.4 ± 8.6
Asym-AO1 [9]		71.2	32.9	19.1	76.3 ± 8.9
BSSF [34]		73.5	39.1	34.2	68.9 ± 8.5
Farsight [16]		73.2	37.8	31.3	72.4 ± 8.6
QME [43]		73.8	39.6	35.0	64.3 ± 8.0
FusionAgent (DA)		**75.5**	**41.0**	36.5	50.3 ± 9.0
FusionAgent (CoT)		**75.5**	**41.0**	**37.0**	$\mathbf{50.0 \pm 8.5}$

Table 2. **Performance on LTCC.** [Keys: **Best** and second best performance; *Comb.*: model combination; *: zero-shot performance; ♦: AdaFace for face modality; ♠: CAL of body modality; ■: AIM for body modality; ●: SapiensID for face and body modalities; TAR: TAR@1%FAR; FNIR: FNIR@1%FPIR.]

$k = 10$ for CCVID and $k = 40$ for MEVID and LTCC. During training, we sample a continuous 4-frame clip from each video for CCVID and MEVID, and 1 frame for LTCC as it is an image-based dataset. Training is conducted on 4 H100 GPUs with an effective batch size of 4, and takes nearly 4 hours. During evaluation, we disable sampling in FusionAgent to ensure results are reproducible.

4.1. Experimental Results

In-domain Evaluation. As shown in Tab. 1, 2, and 3, statistical baselines (*e.g.*, Z-score, FarSight) often fail to surpass the strongest single model across all metrics, while learning-based methods (*e.g.*, QME) achieve better gains and occasionally outperform the best unimodal model, but remain limited when fused models are not complementary (*e.g.*, LTCC). By contrast, FusionAgent consistently achieves superior performance on all three datasets, outperforming both unimodal and fusion baselines on most metrics. A consistent trend across the tables is that FNIR benefits the most from fusion while Rank-1 improves the least: this can be explained because FNIR is particularly sensitive to outliers, whereas Rank-1 is largely determined by the strongest modality and only marginally affected by fusion. Through top-k selection and confidence weighting, our method effectively bounds the increase in non-match scores. On CCVID, the largest TAR arise as FusionAgent identifies the high quality of facial inputs and primarily anchors on the FR model, which excels at distinguishing matches from non-matches. On LTCC, FNIR reduction

Method	*Comb.*	Rank1↑	mAP↑	TAR↑	FNIR↓
*AdaFace** [13]	♦	25.0	8.1	5.4	98.8 ± 1.2
CAL [7]	♠	52.5	27.1	34.7	67.8 ± 7.3
AGRL [36]	■	51.9	25.5	30.7	69.4 ± 8.9
Min-Fusion [11]		46.8	21.2	28.0	70.4 ± 8.0
Max-Fusion [11]		33.2	14.9	8.3	97.4 ± 1.6
Z-score [31]		54.1	27.4	30.7	66.5 ± 7.0
Min-max [31]		52.8	24.7	25.0	71.3 ± 6.1
Weigthed-sum [22]		54.1	27.3	30.3	66.3 ± 7.0
Asym-AO1 [9]	♦♠■	52.5	22.9	23.6	71.7 ± 5.8
BSSF [34]		53.5	27.4	30.5	65.9 ± 7.2
Farsight [17]		53.8	25.4	26.6	69.8 ± 6.4
QME [43]		**55.7**	28.2	32.9	64.6 ± 8.2
FusionAgent (DA)		52.5	**28.7**	34.8	60.8 ± 7.3
FusionAgent (CoT)		54.7	**28.7**	**34.9**	$\mathbf{58.6 \pm 7.4}$

Table 3. **Performance on MEVID.** [Keys: **Best** and second best performance; *Comb.*: model combination; *: zero-shot performance; ♦: AdaFace for face modality; ♠: CAL of body modality; ■: AGRL for body modality; TAR: TAR@1%FAR; FNIR: FNIR@1%FPIR.]

Method	Rank1↑	mAP↑	TAR↑	FNIR↓
CCVID → LTCC (Zero-shot)				
FusionAgent (ACT)	60.4	11.9	7.7	60.3 ± 8.6
FusionAgent (Farsight)	68.2	31.7	17.0	81.8 ± 9.6
CCVID → LTCC (10-shot)				
FusionAgent (ACT)	73.6	39.8	34.8	53.5 ± 8.5
MEVID → LTCC (Zero-shot)				
FusionAgent (Farsight)	75.3	42.3	37.7	59.3 ± 9.5
FusionAgent (ACT)	75.3	41.1	36.1	50.1 ± 8.4

Table 4. **Cross-domain Performance on LTCC.** FusionAgent is trained on CCVID with its model combination (Tab. 1) and evaluated on LTCC using LTCC's model combination (Tab. 2). [Keys: TAR: TAR@1%FAR; FNIR: FNIR@1%FPIR.]

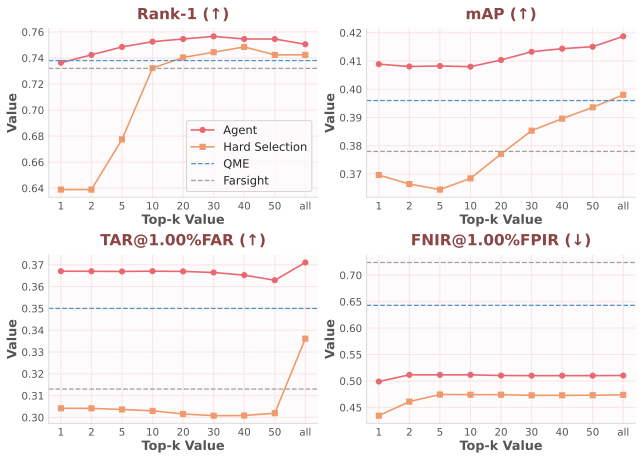

Figure 5. **Performance comparison on LTCC in 4 metrics.** FusionAgent consistently outperforms baselines, including the hard selection (i.e., using all models), which highlights the effectiveness of dynamic model selection.

is most pronounced due to the proposed ACT score-fusion strategy (see ablation in Tab. 5).

Efficiency in Real-world Scenarios. We introduce two inference modes: Direct Answering (DA) and Chain-of-Thought (CoT). DA omits explicit reasoning, reducing average inference time from 2.81s to 1.03s (QME takes 0.67s) while maintaining competitive performance on an H100 device. CoT offers interpretable results, but at the cost of higher latency. Users can flexibly choose between the two to balance efficiency, interpretability, and accuracy.

Cross-domain Evaluation. In real-world deployments, agents might be deployed in unseen environments with different tool combinations, making cross-domain evaluation critical for assessing transferability. In Tab. 4), we evaluate FusionAgent by training on CCVID and testing on LTCC. In the zero-shot setting, although the tool pool is changed, FusionAgent generalizes well and successfully executes tool calls without any additional training. Nevertheless, in CCVID $\rightarrow$ LTCC, where face quality degrades, FusionAgent experiences a performance drop due to (1) the unpredictable behavior of newly introduced models or weights on the target dataset, and (2) the domain gap in data distribution. With only 10 samples per subject (*i.e.*, 8% of the training data) and 50 training steps, FusionAgent rapidly adapts, achieving performance comparable to training on the full in-domain dataset. In MEVID $\rightarrow$ LTCC, the zero-shot setting attains similar performance, further demonstrating the robustness of FusionAgent.

4.2. Ablation Studies

Agent Selection *vs*. Hard Selection. We compare dynamic model selection by the agent (red line) with hard selection, which uses all available models (orange line) in Fig. 5. We use the averaged scores as the anchor for the hard selection. In the agent case, the first selected model serves as the anchor, whereas for hard selection such anchor information is unavailable. Therefore, we use the averaged scores as a surrogate anchor for hard selection. Our method yields notable improvements across all metrics compared to previous SoTA approaches, underscoring the efficacy of the proposed agent and the ACT algorithm. Furthermore, it consistently outperforms hard selection in terms of Rank-1, mAP, and TAR, highlighting the critical role of adaptive model selection. The significantly lower performance of hard selection relative to prior SoTA confirms that dynamic model selection is the key driver of performance gains.

Top-K Values. We investigate the impact of top-k values in ACT. Fig. 5 indicates that the choice of k affects different metrics distinctively: larger k values generally improve Rank-1, mAP, and TAR@FAR, while FNIR remains relatively stable. Compared with hard selection, which exhibits large fluctuations, especially at small k, the proposed agent maintains both higher performance and greater stability across a wide range of k values. This shows that the agent not only adapts to sample-level variations by selectively exploiting complementary models but also avoids the noise introduced by averaging redundant scores. The robustness of ACT under varying k highlights its effectiveness in supporting adaptive model selection, ensuring consistent improvements across diverse evaluation settings.

Alternative Score-fusion Methods. Tab. 5 compares the proposed ACT against alternative statistical score-fusion methods using the same agent-selected model combinations. Even with standard fusion techniques such as Z-score and Farsight, our approach already surpasses the QME, highlighting the decisive role of dynamic model selection. Interestingly, Z-score and Farsight achieve nearly identical results, suggesting that the choice of statistical fusion has a limited impact once adaptive selection is applied. In contrast, the proposed ACT yields further consistent gains, with a substantial reduction in FNIR (down to 51.0). This demonstrates that beyond dynamic selection, robust score integration is crucial for handling challenging open-set search scenarios, making ACT more reliable and generalizable across diverse evaluation metrics.

Statistics of Dynamic Model Selection. Fig. 6 presents the frequency of model and anchor selections per dataset. On CCVID, where faces are often clearly visible, the agent frequently selects AdaFace as the anchor and consistently combines it with CAL, while BigGait is selected less often, suggesting limited complementary value for combination. In contrast, LTCC and MEVID—collected under surveillance conditions with multi-view and low-quality faces—lead the agent to rely on ReID models.

Dataset-level Model Selection. Since exhaustive sample-

Method	Rank1	mAP	TAR@1%FAR	FNIR@1%FPIR
QME [43]	73.8	39.6	35.0	64.3 ± 8.0
Z-score	74.8	**41.7**	37.1	63.7 ± 9.5
Farsight	74.8	**41.7**	**37.2**	62.5 ± 9.7
ACT (Ours)	**75.5**	41.4	36.5	$\mathbf{51.0 \pm 9.4}$

Table 5. **Comparison of score-fusion methods with agent selection on LTCC.** All fusion methods combined with agent-based selection outperform QME, confirming the value of dynamic model selection. The proposed ACT yields the best overall performance, particularly in open-set search.

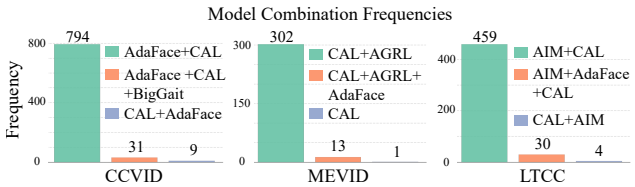

Figure 6. **Statistics of model selection.** The first model is the anchor model for each category. The combination distribution reveals the characteristics of each dataset and which model combination provides robust performance.

Comb.	Rank1	mAP	TAR@1%FAR	FNIR@1%FPIR
AIM+CAL	74.8	41.0	36.1	53.4 ± 9.2
FusionAgent	75.5	41.4	36.5	51.0 ± 9.4

Table 6. **Ablation on model combination and anchor model selection on LTCC.** The results show that FusionAgent closely aligns with the grid-searched results but achieves slightly better performance due to sample-level diversity.

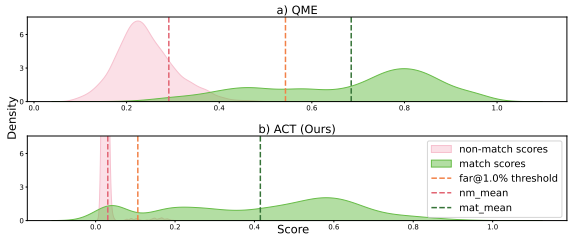

Figure 7. **Comparison of score distribution on CCVID.** [Keys: nm mean = mean value of non-match scores; mat mean = mean; value of match scores.]

level grid search is computationally prohibitive, we conduct grid search at the dataset level to determine the optimal model combination and anchor model. As shown in Tab. 6, AIM+CAL is the best model combination using ACT for the grid search, which is highly consistent with the combination chosen by our agent. Nevertheless, the agent achieves superior performance, attributed to its ability to exploit sample-level diversity during model selection.

4.3. Qualitative Results

Score Distribution. Fig. 7 plots the distribution of non-match (imposter) scores, match (genuine) scores, and the threshold for FAR=1% for QME and FusionAgent. For a better visualization, we align the y-axis and normalize the scores. Our method demonstrates a markedly larger margin between the match scores and the FAR threshold, while effectively suppressing non-match scores near zero. This improvement stems from two key mechanisms: (1) by summing only the top-k confident scores, we amplify the contribution of true matches and widen the separation from non-matches, which are predominantly excluded; and (2) our dynamic model selection provides a more robust set of models per query, leading to higher-quality scores for fusion.

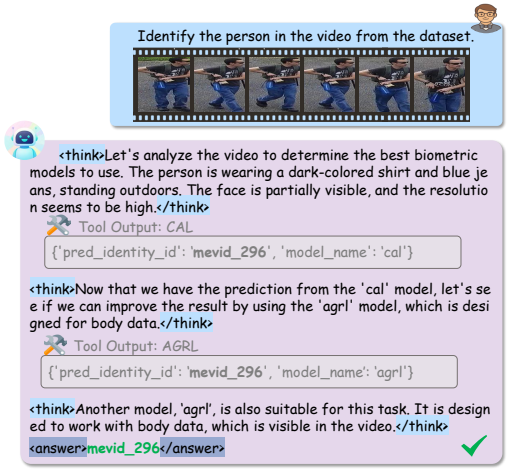

Figure 8. **CoT of FusionAgent on MEVID.** FusionAgent performs interpretable reasoning and dynamically selects the suitable tool combination for each test sample.

CoT for Model Selection. Fig. 8 illustrates the CoT mode of FusionAgent, where the agent interprets the input and dynamically selects suitable models for each query sample. By leveraging the learned reliability and complementarity of different models, FusionAgent reasons over the query's characteristics to determine the optimal model combination. This CoT process enhances interpretability by explicitly revealing the agent's decision path and selection rationale.

5. Conclusion

We propose FusionAgent, a novel agentic framework for human recognition with dynamic model selection. We introduce an Anchor-based Confidence Top-k score-fusion method (ACT) for sample-dependent and adaptive model integration. Multiple reward functions are designed to guide the agent in tool-use and exploration across diverse model combinations. Extensive experiments and analyses validate the effectiveness of FusionAgent and ACT, highlighting the benefit of query-wise model selection and fusion. Our approach provides a scalable and extensible solution for multi-modal and multi-model tasks, with potential applicability to broader scenarios such as vision-language retrieval.

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